WHAT'S THE "ANTI"MATTER! ANTINEUTRON SIGNALS AT CLAS12

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Motivation

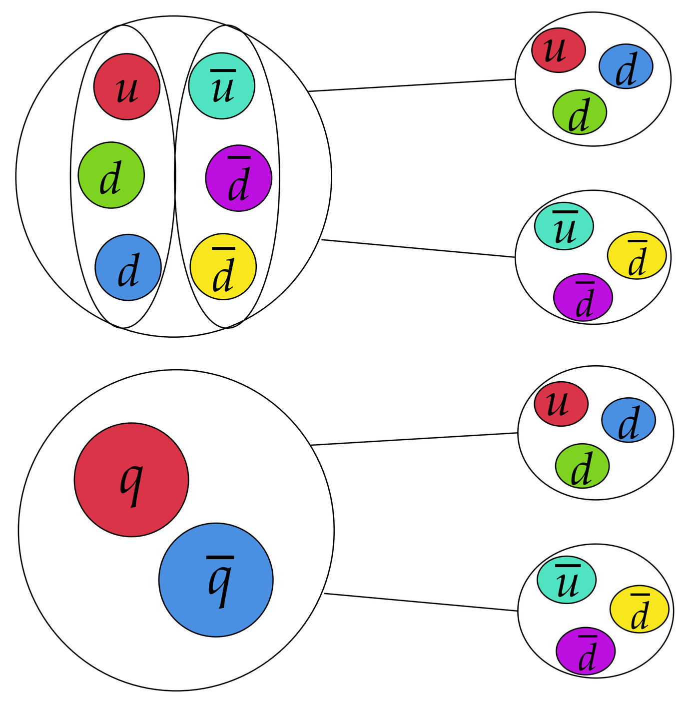
Why Antineutron Electroproduction?
Antimatter: Antineutrons are the least understood building block of antimatter


Positron & Antiproton Research


Antineutron Research

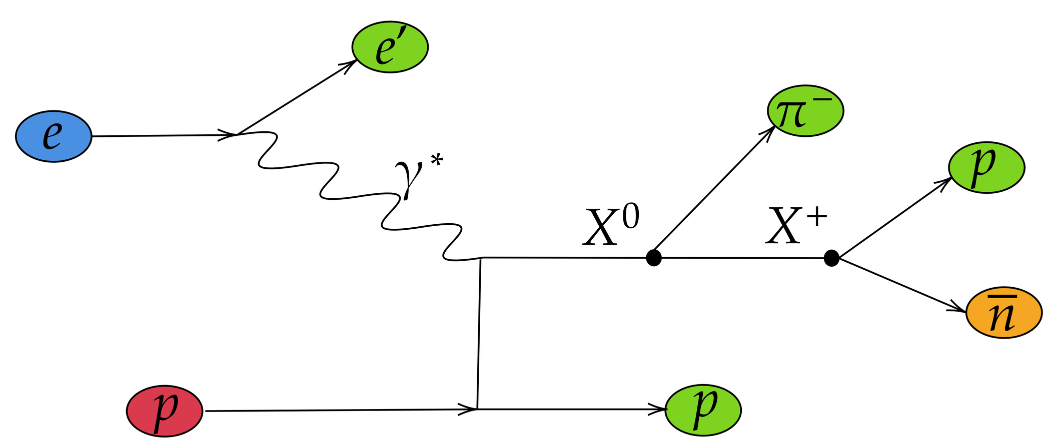
Exclusivity: Electroproduction offers a uniquely clean, precise method to isolate and study their properties
Sparse Data: This experiment will extract one of the *first ever* cross-section measurements for $\bar{n}$ electroproduction
Potential Exotic States:

- Baryonium bound system
- Multiquark: e.g. Bound tetra- or pentaquark state
- Glueballs
- Intermediate Mesons



Mechanism

Antineutron is indirectly detected using conservation of energy and momentum:



References

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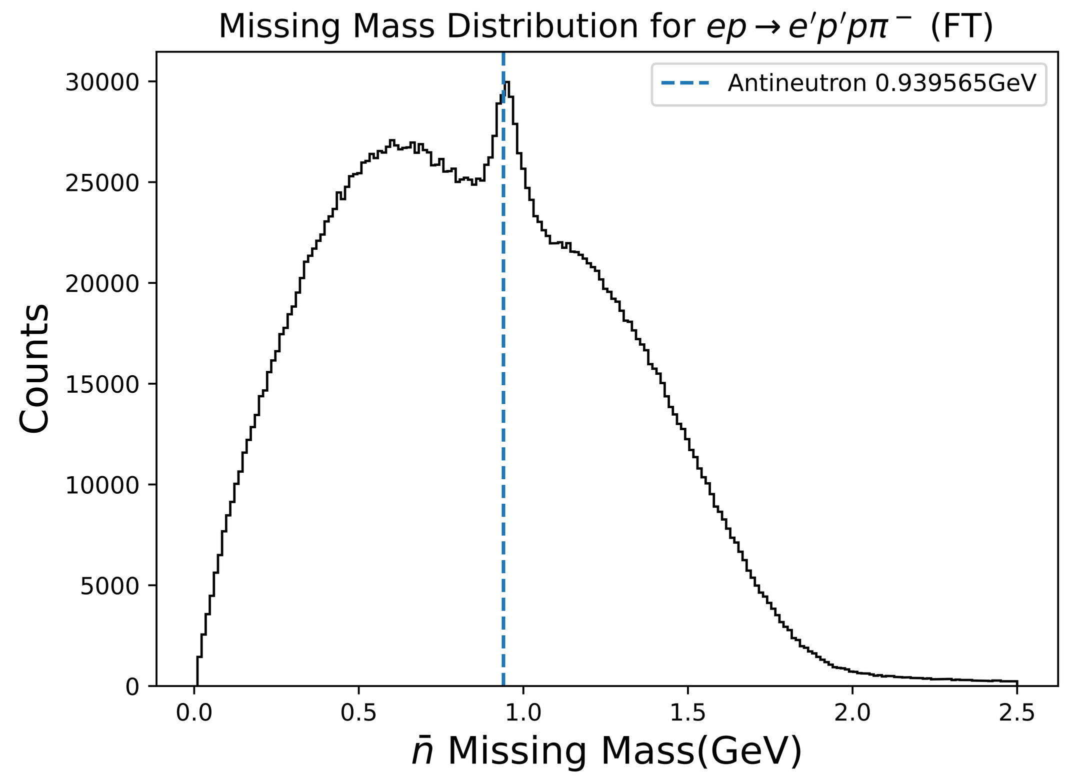
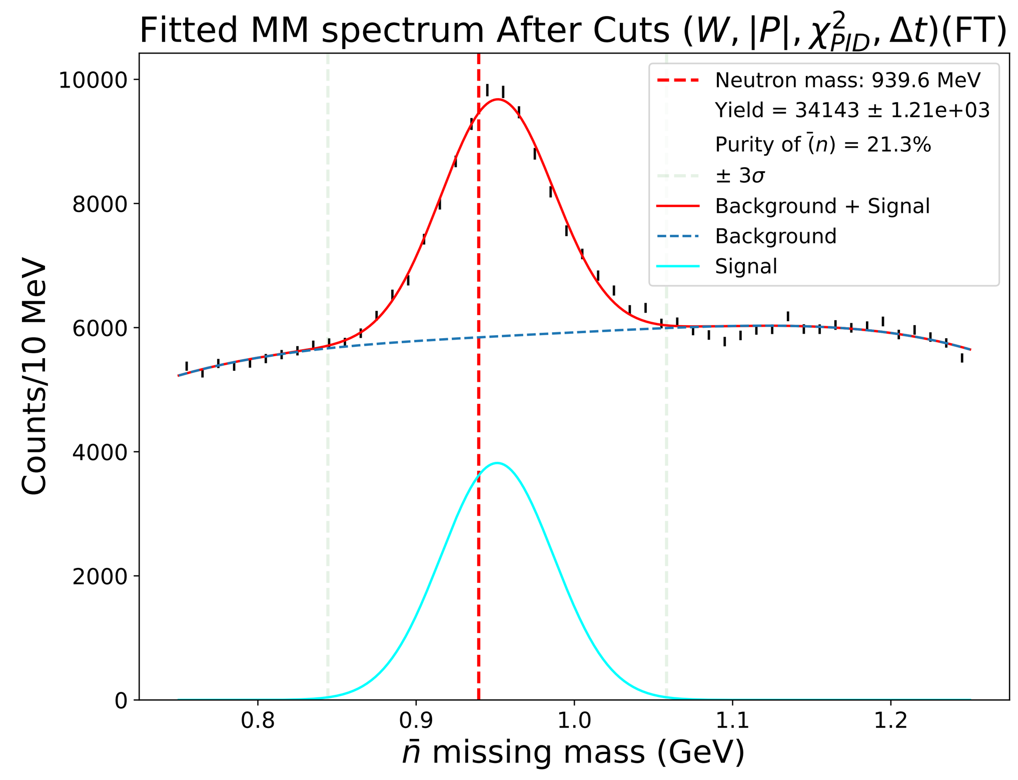
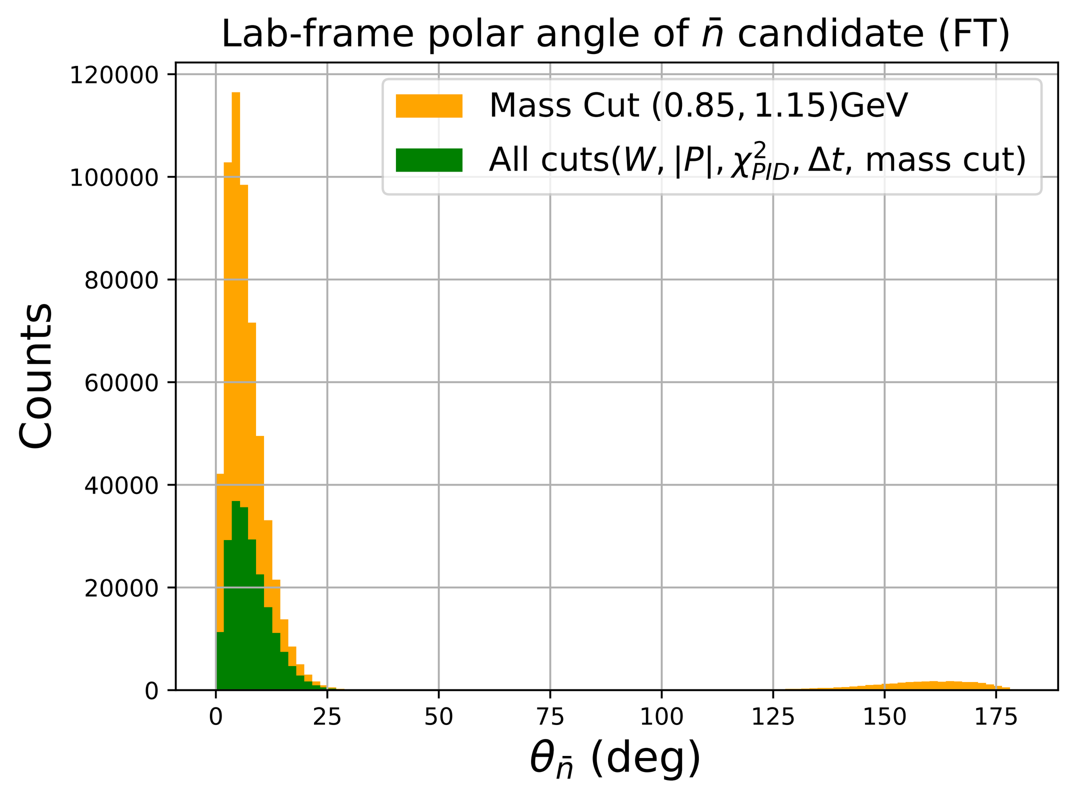
Forward Tagger Data



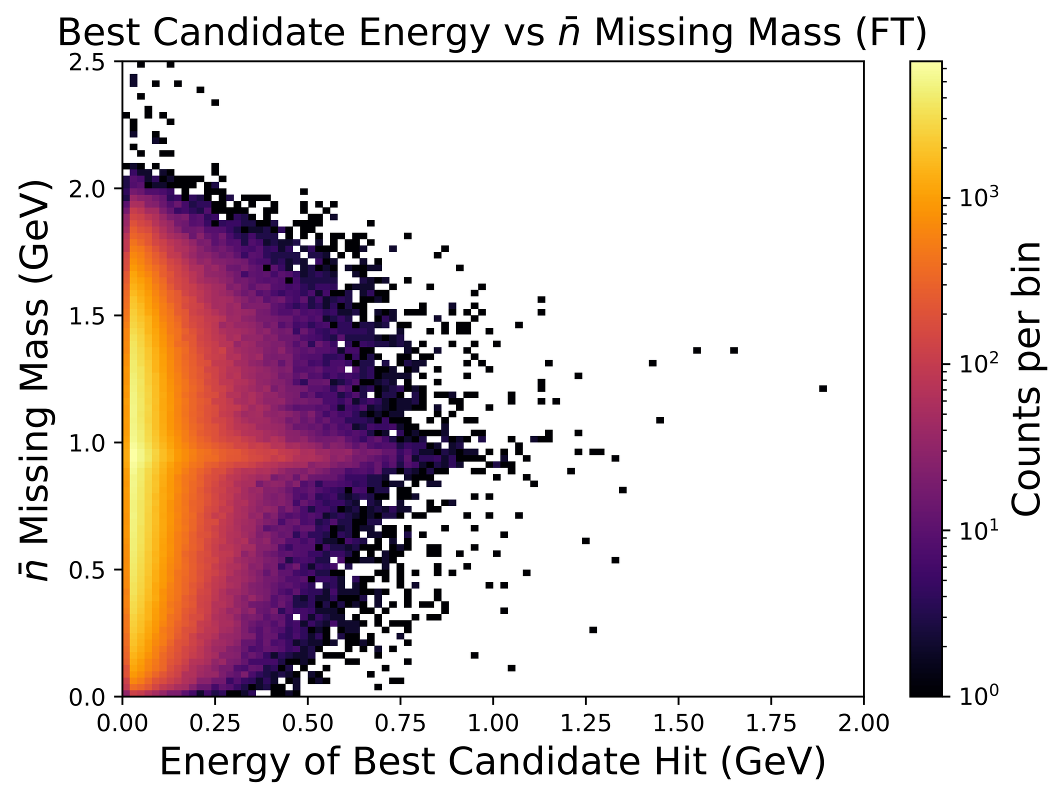
Raw Spectrum: Showing the broad background and the expected $\bar{n}$ peak near 0.94 GeV that guides the later selection.



Signal Extraction: Successful isolation of the $\bar{n}$ and applied fit (Gaussian + 4^{th} order polynomial) allows us to extract the total event yield

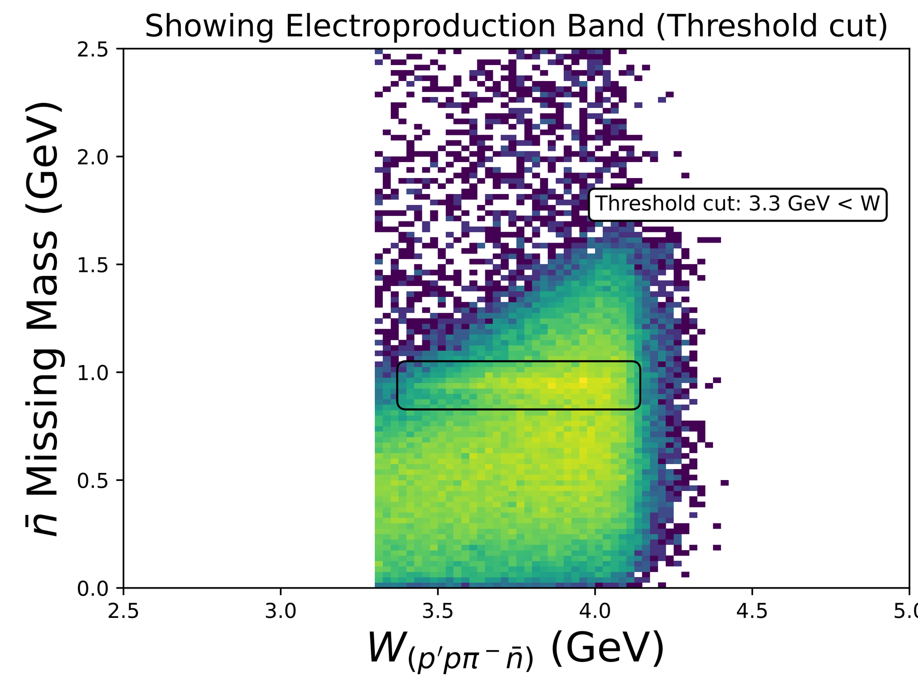


Detector Acceptance: Lab-frame polar angle distribution illustrating where the detector has highest sensitivity to the signal.

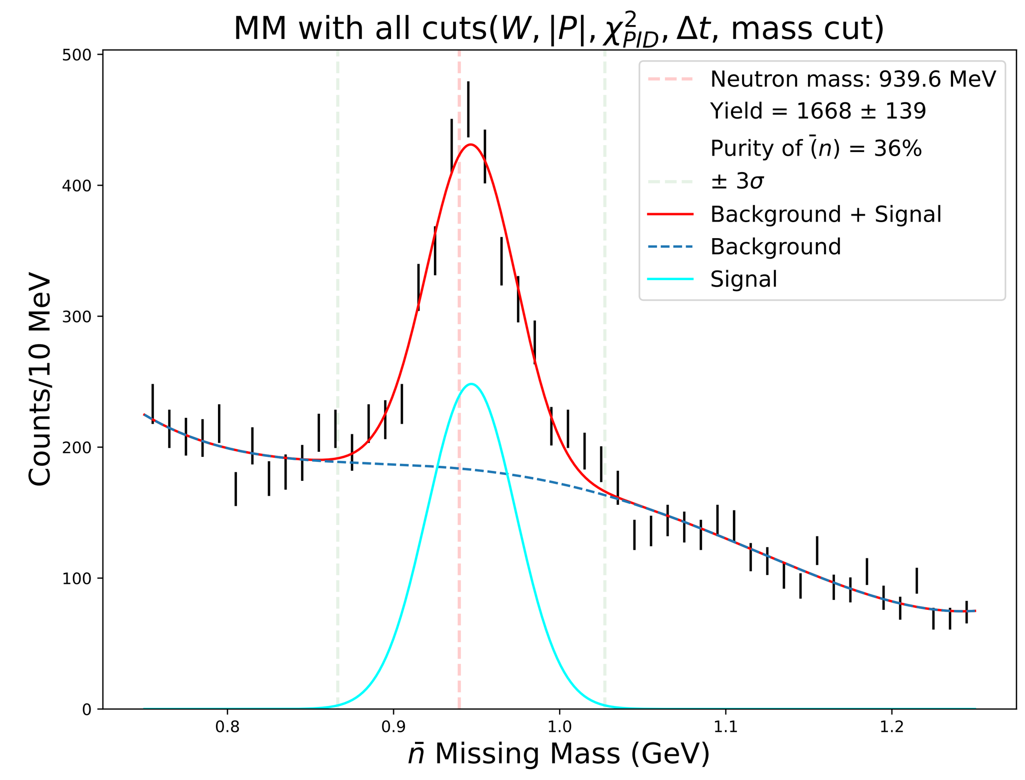


Candidate Energy Spectrum: Measured energy deposition of the isolated $\bar{n}$ candidates in ECAL.

Forward Detector Data

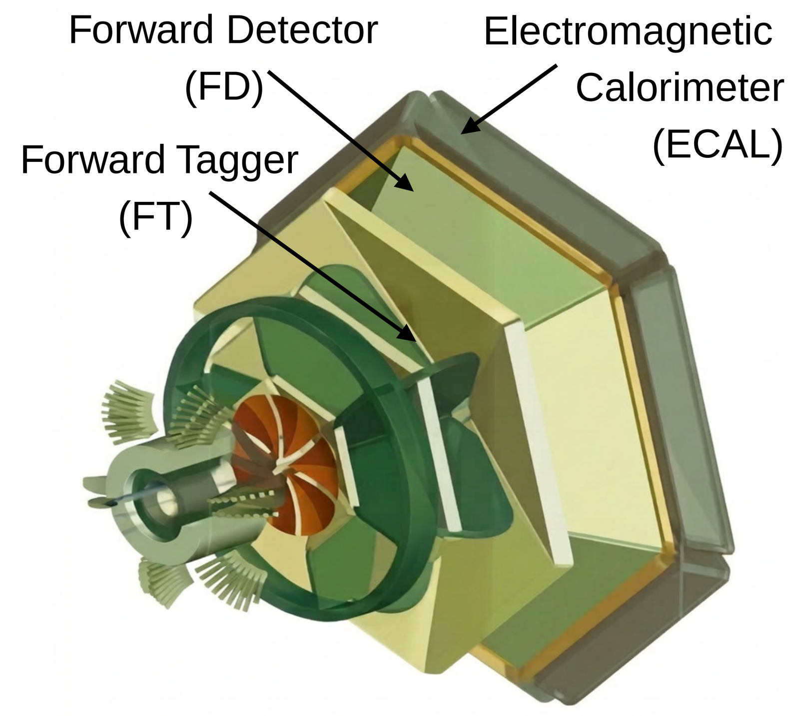


Reaction Kinematics: Shows the production threshold and helps identify where the background is most dominant



Signal Extraction: Successful isolation of the $\bar{n}$ and applied fit

CLAS12



- The CLAS12 spectrometer provides large acceptance and high resolution ($\sim 80 - 150ps$)
- Operates at a high luminosity of $10^{35}\ cm^{-2}\ s^{-1}$

Outlook

- Estimate the background through data driven techniques

N	CLAS6(γ) Counts	CLAS12 Counts
# $\bar{p}$	229,117	350-400K
# $\bar{n}$	1,200	35,811

Acknowledgments

This work was supported by the US Department of Energy Office of Science, Office of Nuclear Physics, under contract No. DE-SC0022007